Assignment 1

Mech 587

Due: Never; this isn't a real assignment.

1. (10 marks) At boundaries, centered flux approximations are possible only through the use of ghost cells. Suppose that you wanted a second-order accurate approximation of the Poisson flux $\frac{\partial T}{\partial x}$ at the right boundary of a one-dimensional domain, using only interior control volume values.

   (a) How many control volumes will you need to use in this approximation?

   When you write the control volume averages as series expansions and sum, you need to match three terms: T, $\frac{\partial T}{\partial x}$, and $\frac{\partial^2 T}{\partial x^2}$. To do this requires three control volumes, unless you get some fortuitous cancelation.

   (b) What will the flux be? Write your answer in the form

$$\left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2}} = a\overline{T}_i + b\overline{T}_{i-1} + c\overline{T}_{i-2} + \cdots,$$

   using as many terms as required for second-order accuracy.

   We already know (from class notes) what the control volume averages are for i and $i-1$. For $i-2$, the result is:

$$\begin{aligned}
\overline{T_{i-2}} &= \frac{1}{\Delta x}\int_{x_{i-\frac{5}{2}}}^{x_{i-\frac{3}{2}}} T(x)\,dx \\
&= \frac{1}{\Delta x}\left[Tx + T'\frac{x^2}{2} + T''\frac{x^3}{6} + T'''\frac{x^4}{24} + \cdots\right]_{-3\Delta x}^{-2\Delta x} \\
&= \frac{1}{\Delta x}\left[T\,\Delta x - T'\frac{5\Delta x^2}{2} + T''\frac{19\Delta x^3}{6} - T'''\frac{65\Delta x^4}{24} + \cdots\right] \\
&= T - T'\frac{5\Delta x}{2} + T''\frac{19\Delta x^2}{6} - T'''\frac{65\Delta x^3}{24} + \cdots
\end{aligned}$$

   Now I'll switch to using T, and write a Taylor table:

	T	$\frac{\partial T}{\partial x}$	$\frac{\partial^2 T}{\partial x^2}$	$\frac{\partial^3 T}{\partial x^3}$
$\overline{T}_i$	1	$-\frac{\Delta x}{2}$	$\frac{\Delta x^2}{6}$	$-\frac{\Delta x^3}{24}$
$\overline{T}_{i-1}$	1	$-\frac{3\Delta x}{2}$	$\frac{7\Delta x^2}{6}$	$-\frac{15\Delta x^3}{24}$
$\overline{T}_{i-2}$	1	$-\frac{5\Delta x}{2}$	$\frac{19\Delta x^2}{6}$	$-\frac{65\Delta x^3}{24}$
$a\overline{T}_i + b\overline{T}_{i-1} + c\overline{T}_{i-2}$	$a+b+c$	$-\frac{\Delta x}{2}(a+3b+5c)$	$\frac{\Delta x^2}{6}(a+7b+19c)$	$-\frac{\Delta x^3}{24}(a+15b+65c)$
Goal: $\frac{\partial T}{\partial x}$	0	1	0	???

   So we need to solve the following linear system

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 5 \\ 1 & 7 & 19 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{2}{\Delta x} \\ 0 \end{pmatrix}$$

   So that gives me $c = \frac{1}{\Delta x}$, $b = -\frac{3}{\Delta x}$, $a = \frac{2}{\Delta x}$. Finally, we get (evaluating the leading-order error term):

$$\frac{2\overline{T}_i - 3\overline{T}_{i-1} + \overline{T}_{i-2}}{\Delta x} = \left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2}} - \frac{11}{12}\left.\frac{\partial^3 T}{\partial x^3}\right|_{i+\frac{1}{2}}\Delta x^3 + O\left(\Delta x^4\right)$$

   This can be re-written as:

$$\frac{2\overline{T}_i - 3\overline{T}_{i-1} + \overline{T}_{i-2}}{\Delta x} = 3\frac{\overline{T}_i - \overline{T}_{i-1}}{\Delta x} - 2\frac{\overline{T}_i - \overline{T}_{i-2}}{2\Delta x}$$

where each of the terms on the right is a first derivative approximation (first-order accurate at $i+\frac{1}{2}$); combining the two appropriately cancels their error to one order higher.

2. (10 marks) Suppose that we want a flux approximation for the Laplacian that is more than second-order accurate — for example, fourth-order accurate.[1] In this problem, you will use the control volume averages in control volumes $i-1$, i, $i+1$, and $i+2$ to construct a fourth-order accurate flux approximation at $i+\frac{1}{2}$. *I strongly encourage you to use Maple, or some other symbolic math software for this problem! If nothing else, it'll give you a chance to get acquainted with the software on a (relatively) easy problem before we get to harder things later on.*

   (a) Expand a function $f(x)$ in a Taylor series around $x_{i+\frac{1}{2}}$. (It is important to expand the function about this location so that the sum of Taylor series will reflect accuracy relative to the point $i+\frac{1}{2}$.)

$$\begin{aligned} f(x) &= f\left(x_{i+\frac{1}{2}}\right) + \frac{df\left(x_{i+\frac{1}{2}}\right)}{dx}\left(x - x_{i+\frac{1}{2}}\right) + \frac{d^2f\left(x_{i+\frac{1}{2}}\right)}{dx^2}\frac{\left(x - x_{i+\frac{1}{2}}\right)^2}{2} \\ &\quad + \frac{d^3f\left(x_{i+\frac{1}{2}}\right)}{dx^3}\frac{\left(x - x_{i+\frac{1}{2}}\right)^3}{6} + \frac{d^4f\left(x_{i+\frac{1}{2}}\right)}{dx^4}\frac{\left(x - x_{i+\frac{1}{2}}\right)^4}{24} \\ &\quad + \frac{d^5f\left(x_{i+\frac{1}{2}}\right)}{dx^5}\frac{\left(x - x_{i+\frac{1}{2}}\right)^5}{120}\cdots \end{aligned}$$

   (b) Use the expansion from part (a) to compute the control volume averages of $f(x)$ in each of the control volumes $i-1$, i, $i+1$, and $i+2$.

   For control volume $i+1$, we have (dropping the locations at which derivatives are evaluated to save space; all are evaluated at $x_{i+\frac{1}{2}}$)

$$\begin{aligned} \overline{f_{i+1}} &= \frac{1}{\Delta x}\int_{x_{i+\frac{1}{2}}}^{x_{i+\frac{3}{2}}} f(x)\,dx \\ &= \frac{1}{\Delta x}\left[fx + f'\frac{x^2}{2} + f''\frac{x^3}{6} + f'''\frac{x^4}{24} + f''''\frac{x^5}{120} + f'''''\frac{x^6}{720} + \cdots\right]_0^{\Delta x} \\ &= \frac{1}{\Delta x}\left[f\,\Delta x + f'\frac{\Delta x^2}{2} + f''\frac{\Delta x^3}{6} + f'''\frac{\Delta x^4}{24} + f''''\frac{\Delta x^5}{120} + f'''''\frac{\Delta x^6}{720} + \cdots\right] \\ &= f + f'\frac{\Delta x}{2} + f''\frac{\Delta x^2}{6} + f'''\frac{\Delta x^3}{24} + f''''\frac{\Delta x^4}{120} + f'''''\frac{\Delta x^5}{720} + \cdots \end{aligned}$$

   By inspection, we can write the average for control volume i as well:

$$\overline{f}_i = f - f'\frac{\Delta x}{2} + f''\frac{\Delta x^2}{6} - f'''\frac{\Delta x^3}{24} + f''''\frac{\Delta x^4}{120} - f'''''\frac{\Delta x^5}{720} + \cdots$$

   Repeating the process for control volume $i+2$,

$$\begin{aligned} \overline{f_{i+2}} &= \frac{1}{\Delta x}\int_{x_{i+\frac{3}{2}}}^{x_{i+\frac{5}{2}}} f(x)\,dx \\ &= \frac{1}{\Delta x}\left[fx + f'\frac{x^2}{2} + f''\frac{x^3}{6} + f'''\frac{x^4}{24} + f''''\frac{x^5}{120} + f'''''\frac{x^6}{720} + \cdots\right]_{\Delta x}^{2\Delta x} \\ &= \frac{1}{\Delta x}\left[f\,\Delta x + f'\frac{3\Delta x^2}{2} + f''\frac{7\Delta x^3}{6} + f'''\frac{15\Delta x^4}{24} + f''''\frac{31\Delta x^5}{120} + f'''''\frac{63\Delta x^6}{720} + \cdots\right] \\ &= f + f'\frac{3\Delta x}{2} + f''\frac{7\Delta x^2}{6} + f'''\frac{5\Delta x^3}{8} + f''''\frac{31\Delta x^4}{120} + f'''''\frac{7\Delta x^5}{80} + \cdots \end{aligned}$$

[1] Even orders of accuracy turn out to require no more points than odd orders for the Laplacian operator on uniform meshes, so third-order accuracy isn't worth the trouble.

	f	$f'\Delta x$	$f''\frac{\Delta x^2}{2}$	$f'''\frac{\Delta x^3}{6}$	$f''''\frac{\Delta x^4}{24}$	$f'''''\frac{\Delta x^5}{120}$
$a\overline{f_{i+2}}$	a	$a\frac{3}{2}$	$a\frac{7}{3}$	$a\frac{15}{4}$	$a\frac{31}{5}$	$a\frac{21}{2}$
$b\overline{f_{i+1}}$	b	$\frac{b}{2}$	$\frac{b}{3}$	$\frac{b}{4}$	$\frac{b}{5}$	$\frac{b}{6}$
$c\overline{f_i}$	c	$-\frac{c}{2}$	$\frac{c}{3}$	$-\frac{c}{4}$	$\frac{c}{5}$	$-\frac{c}{6}$
$d\overline{f_{i-1}}$	d	$-d\frac{3}{2}$	$d\frac{7}{3}$	$-d\frac{15}{4}$	$d\frac{31}{5}$	$-d\frac{21}{2}$
Sum	$(a+d)+(b+c)$	$\frac{3(a-d)+(b-c)}{2}$	$\frac{7(a+d)+(b+c)}{3}$	$\frac{15(a-d)+(b-c)}{4}$	$\frac{31(a+d)+(b+c)}{5}$	$\frac{63(a-d)+(b-c)}{6}$
Goal	0	$\frac{1}{\Delta x}$	0	0	0	?

Table 1: Taylor table for fourth-order accurate approximation to the Laplace flux.

Finally, by inspection we can the average for control volume $i-1$:

$$\overline{f_{i-1}} = f - f'\frac{3\Delta x}{2} + f''\frac{7\Delta x^2}{6} - f'''\frac{5\Delta x^3}{8} + f''''\frac{31\Delta x^4}{120} - f'''''\frac{7\Delta x^5}{80} + \cdots$$

(c) Find a linear combination of this control volume averages which gives a fourth-order accurate approximation to the first derivative at $x_{i+\frac{1}{2}}$. What is the leading-order term in the truncation error in this expression?

To match the first derivative to fourth-order accuracy, we need to have a combination that gives the first-derivative with a coefficient of one and zeroth, second, third and fourth derivatives with a coefficient of zero. All the expansions are written in a Taylor table with unknown coefficients in Table 1. Solving the resulting set of linear equations gives the following approximation to the first derivative:

$$\frac{-\overline{f_{i+2}} + 15\overline{f_{i+1}} - 15\overline{f_i} + \overline{f_{i-1}}}{12\Delta x} = \frac{df}{dx}_{i+\frac{1}{2}} - \frac{\Delta x^4}{90}\frac{d^5 f}{dx^5}_{i+\frac{1}{2}} + O\left(\Delta x^6\right)$$

The leading-order term in the truncation error is $-\frac{\Delta x^4}{90}\frac{d^5 f}{dx^5}_{i+\frac{1}{2}}$.

(d) The flux at $x_{i-\frac{1}{2}}$ can be obtained by inspection using an index shift; what is it? What will be the leading-order term in its truncation error?

$$\frac{-\overline{f_{i+1}} + 15\overline{f_i} - 15\overline{f_{i-1}} + \overline{f_{i-2}}}{12\Delta x} = \frac{df}{dx}_{i-\frac{1}{2}} - \frac{\Delta x^4}{90}\frac{d^5 f}{dx^5}_{i-\frac{1}{2}} + O\left(\Delta x^6\right)$$

The leading-order term in the truncation error is $-\frac{\Delta x^4}{90}\frac{d^5 f}{dx^5}_{i-\frac{1}{2}}$.

(e) Perform the flux integral for control volume i to obtain a fourth-order accurate finite-volume approximation to the Laplacian.

The flux integral is easily assembled from the fluxes:

$$\begin{aligned}\oint_{CV} \vec{F}\cdot\hat{n}\,ds &= \frac{-\overline{f_{i+2}} + 15\overline{f_{i+1}} - 15\overline{f_i} + \overline{f_{i-1}}}{12\Delta x} \\ &\quad - \frac{-\overline{f_{i+1}} + 15\overline{f_i} - 15\overline{f_{i-1}} + \overline{f_{i-2}}}{12\Delta x} \\ &= \frac{-\overline{f_{i+2}} + 16\overline{f_{i+1}} - 30\overline{f_i} + 16\overline{f_{i-1}} - \overline{f_{i-2}}}{12\Delta x}\end{aligned}$$

Recall that this is used in the context of either

$$\oint_{CV} \vec{F}\cdot\hat{n}\,ds = \bar{S}\,\Delta x$$

or

$$\frac{1}{\Delta x}\oint_{CV} \vec{F}\cdot\hat{n}\,ds = \bar{S}$$

In the latter case, the flux integral divided by Δx looks comfortingly like a second difference term. In fact, this expression can be broken down into:

$$\frac{-\overline{f_{i+2}} + 16\overline{f_{i+1}} - 30\overline{f_i} + 16\overline{f_{i-1}} - \overline{f_{i-2}}}{12\Delta x^2} = \frac{4}{3}\frac{\overline{f_{i+1}} - 2\overline{f_i} + \overline{f_{i-1}}}{\Delta x^2} - \frac{1}{3}\frac{\overline{f_{i+2}} - 2\overline{f_i} - \overline{f_{i-2}}}{4\Delta x^2}$$

where both large fractions on the right-hand side are second-order accurate second-derivative approximations. This combination of the two eliminates — for finite volume methods — the second-order error term as well. Since the approximations are symmetric, no third-order term ever appears, and the result is fourth-order accurate.